

Presentation Script — AI in FinTech: Credit Scorecard System

Total duration: ~25–30 minutes (6 speakers × ~4–5 minutes each)

Tools: MATLAB Credit Scorecard Toolbox · `DataProjScoreCard.xlsx` · **RiskLens** dashboard

Before presenting: Open the RiskLens dashboard (`cd fintech-dashboard → npm run dev → http://localhost:5173/`).

Keep the written report (`Final/fintech (1).pdf`) open as backup on a second device.

Timing overview

Speaker	Section	Target time
1	Introduction	4–5 min
2	Problem / Background	4–5 min
3	Methodology (Part 1)	4–5 min
4	Methodology (Part 2) + Validation	4–5 min
5	Results / Analysis	4–5 min
6	Discussion, Conclusion & Q&A handoff	4–5 min

Slide map (RiskLens navbar order)

Use the **numbered sidebar** in the dashboard — steps **1–11** match this table. PowerPoint title slides are separate; live demo slides start at **1**.

Step	Dashboard page	Slide name (say aloud)	Primary speaker
1	Overview (/)	Introduction & Problem / Data	Speakers 1, 2, 3 (workflow recap)
2	WOE & Binning (/woe)	WOE & Binning Analysis	Speaker 3
3	Scorecard (/scorecard)	Credit Scorecard Points	Speaker 3
4	ROC & Validation (/validation)	ROC Curve & Validation Metrics	Speaker 4
5	Portfolio Risk (/portfolio)	Portfolio Scoring — Score vs PD	Speaker 4
6	Accept / Reject (/decisions)	Accept vs Reject Decisions	Speaker 5
7	Expected Loss (/expected-loss)	Expected Loss Dashboard	Speaker 5
8	Risk Pricing (/pricing)	Risk-Based Interest Rates	Speaker 5
9	Model Performance (/performance)	Model Performance Summary	Speaker 6
10	Final Decision (/final)	Final Portfolio Decision	Speaker 6
11	Live Client Test (/test)	Live Client Test (Demo)	Speaker 6
12	MATLAB & Excel Charts (/outputs/matlab)	MATLAB & Excel Source Charts <i>(appendix)</i>	Speaker 2 / 5 <i>(optional)</i>

Client Explorer (/explorer) is not numbered — use only if asked in Q&A.

Speaker 1 — Introduction

~4–5 minutes

[Slide: Title — “AI in FinTech: Credit Scorecard & Portfolio Optimization”]

Good morning / good afternoon everyone. Today’s presentation is **Portfolio Optimisation and Pricing Interest Rate by Using a Scoring Model** — a complete credit risk workflow built in **MATLAB** and visualized in our **RiskLens** dashboard.

[Pause for 2 seconds. Make eye contact with the audience.]

The core idea is straightforward: a bank receives loan applications and must decide three things for each client — **should we lend**, **how much could we lose if they default**, and **what is the minimum interest rate** that covers that expected loss. We answer these questions using a **credit scorecard** — a statistical model that converts borrower profile data into a score from **0 to 100**, where a higher score means lower risk.

[Switch to Slide 1 — Introduction & Problem / Data: RiskLens Overview (/)]

RiskLens is the interactive front end of our project. Every figure and metric on screen is tied to the same MATLAB pipeline and final report — think of it as a live control room for the scorecard, not a separate set of numbers.

[Point at the four metric cards at the top of the Overview page.]

At a glance, our project works with **500 historical clients** for training and validation, achieves an **AUC of 0.6775**, and when applied to **20 portfolio applicants**, accepts **14 clients** with a total expected loss of **\$129,329.63** on **\$1.4 million** of exposure.

[Point at the workflow pipeline section on the Overview page.]

The workflow follows ten clear steps: load data from Excel, build the scorecard on all 500 historical records, run autobinning and Weight of Evidence analysis, fit a logistic regression model, scale scores to 0–100, validate with a ROC curve, score the portfolio, apply an accept/reject rule, compute expected loss, and finally calculate risk-based minimum interest rates.

[Emphasize this result.]

One design choice worth stating up front: we train on the **full 500 historical clients** — there is **no separate hold-out test sample** — and we validate model performance on that same **HistoricalData** set, as specified in the project requirements.

[Point at the workflow cards again — trace left to right with your hand.]

The presentation follows that same order: **background and data**, **methodology and validation**, **portfolio results in dollars**, and finally **interpretation and conclusions**. Each segment is about four to five minutes, and we will leave time for questions at the end.

[Pause for 2 seconds.]

Next, we move from this overview into the **business problem and the data** behind the model.

Transition → Speaker 2

[Brief handoff, optional:]

“Let’s look at why credit scoring matters and what data we used.”

Speaker 2 — Problem / Background

~4–5 minutes

[Slide 1 — Introduction & Problem / Data (continued)]

Building on that overview, this section covers the **problem background** and **exploratory data analysis**. Stay on **Slide 1 — Overview** in RiskLens.

When a bank receives a loan application, it faces a simple but costly question: **Will this client repay, or will they default?** If the bank accepts too many risky clients, it loses money. If it rejects too many safe clients, it loses business. A **credit scorecard** helps the bank rank applicants from safer to riskier using observable characteristics — age, income, housing status, and employment.

[Slide 1 — Overview: scroll to Default Distribution]

[Point at the pie chart on the left — Good vs Bad.]

Our historical dataset contains **500 clients** from the **HistoricalData** sheet in `DataProjScoreCard.xlsx`. Each record includes **Age**, **Income**, **ResidentialStatus**, **EmploymentStatus**, and a **Default** flag — where **0 means good** (no default) and **1 means bad** (default).

[Point at the numbers beside the pie chart.]

The historical default rate is **36%**: **320 good clients** and **180 bad clients**. That means more than one in three historical borrowers defaulted — a substantial level of credit risk that makes modeling worthwhile.

[Point at the “Default rate by income band” bar chart.]

When we explore the data, a clear pattern appears. **Lower income bands show higher default rates**. For example, clients earning under **\$30,000** have a default rate of about **56.7%**, while those above **\$50,000** have about **21.4%**. This tells us that **Income** will likely be one of the strongest predictors in our model — and our final results confirm that.

[Point at the “Default rate by age group” bar chart.]

Age shows a similar story: **younger applicants** tend to default more often than **older applicants**. Homeowners default less frequently than renters, and employed clients default less than those in the “Other” employment category.

[Slide 1 — Overview: scroll to Portfolio applicants section]

Beyond historical data, we also have **ActualPortfolioData**: **20 new applicants** the bank wants to evaluate. Each has the same demographic fields but **no default label** — because they are future clients. Our job is to score them, decide accept or reject, estimate expected loss, and set minimum pricing.

[Emphasize this result.]

Key assumptions used throughout the project:

- **Loan amount (EAD)**: \$100,000 per client
- **Recovery rate**: 60% if default occurs
- **Loss Given Default (LGD)**: 40% (= 1 – recovery rate)

These assumptions flow through every dollar figure in our results.

[Optional: Slide 12 — MATLAB & Excel Source Charts (/outputs/matlab) — historical default distribution]

[Point at the historical default distribution chart.]

This chart in our MATLAB/Excel outputs section confirms the same 320 vs 180 split visually — good clients in green, bad clients in red.

[Pause for 2 seconds.]

That gives us a clear foundation: **500 labeled historical records** for learning patterns, and **20 new applicants** to score and price. The next section walks through **how the scorecard is built** in MATLAB.

Transition → Speaker 3

[Brief handoff:]

“With the data in place, here is how we built the model.”

Speaker 3 — Methodology / Process (Part 1)

~4–5 minutes

[Slide 1 — Overview: 10-step workflow pipeline (*methodology intro*)]

This part covers the **first half of the methodology**: data preparation, binning, Weight of Evidence, and scorecard construction.

All code is in `FinTech_Scorecard_Project.m`, using MATLAB’s **Credit Scorecard Toolbox**.

[Slide 1 — Overview: scroll to the 10-step workflow pipeline]

[Point at Steps 1–5.]

Step 1 — Data loading: We read `HistoricalData` and `ActualPortfolioData` from Excel. We convert categorical variables and remove empty rows from the portfolio sheet.

Step 2 — Scorecard setup: We create a `creditscorecard` object with **Default** as the response variable and **GoodLabel = 0**. Crucially, we train on **all 500 historical clients**, as required by the assignment slide — no out-of-bag split.

Step 3 — Autobinning & WOE: We call `autobinning()` to let MATLAB find **data-driven optimal cut points** for Age and Income, rather than choosing arbitrary bins by hand. For each predictor we run `bininfo()` and `plotbins()` — which the assignment explicitly requires.

[Switch to Slide 2 — WOE & Binning Analysis (/woe)]

[Point at the Information Value summary cards at the top.]

Weight of Evidence — **WOE** — measures how much each bin separates good payers from bad payers. **Positive WOE** means the bin is safer than average; **negative WOE** means it is riskier.

The **Information Value (IV)** ranks predictors:

- **Income: 0.2355 — Strong**
- **Age: 0.1879 — Medium**
- **EmploymentStatus: 0.1143 — Medium**
- **ResidentialStatus: 0.0417 — Weak**

[Point at the main WOE chart — Income selected.]

[Point at the green and red bars above and below the zero line.]

On this chart, income bins below **\$28,000** have strongly **negative WOE** — high risk. Bins above **\$50,000** have **positive WOE** — low risk. The zero reference line makes the interpretation easy: above zero is safer, below zero is riskier.

[Scroll down to the four predictor cards with tables.]

Each card shows the bin, number of observations, default percentage, and WOE value — matching the **bininfo** tables in our written report.

[Switch to Slide 3 — Credit Scorecard Points (/scorecard)]

[Point at the Income bar chart.]

Step 4 — Logistic regression: After WOE transformation, we call `fitmodel()`. All four predictors — Age, Income, ResidentialStatus, and EmploymentStatus — are statistically significant and retained in the model.

Step 5 — Score scaling: We use `formatpoints()` with **WorstAndBestScores [0, 100]** so every applicant receives a score between **0 and 100**. Higher score means **lower risk**.

[Point at the scaled points table for Income.]

For example, the highest income bin receives about **28.91 points**, while the lowest income bin receives **-8.68 points**. Employed clients receive **25.12 points** versus only **1.80** for “Other” employment. Homeowners receive **21.10 points** versus **6.71** for renters.

[Pause for 2 seconds.]

At this point the scorecard is fully defined — but the critical question remains: **does it actually rank risky clients correctly?** The next section answers that with **ROC validation** and the **KS-based cutoff**.

Transition → Speaker 4

[Brief handoff:]

“Next: validating the model and setting the acceptance threshold.”

Speaker 4 — Methodology (Part 2) + Validation / Results Setup

~4–5 minutes

[Slide 4 — ROC Curve & Validation Metrics]

This section covers **model validation**, how the **accept/reject cutoff** is chosen, and how the **20 portfolio applicants** are scored.

[Switch to Slide 4 — ROC & Validation (/validation)]

[Point at the four metric cards.]

After fitting the scorecard, we run `validateModel()` on the **full HistoricalData** — all 500 clients — and plot the **ROC curve**. Here are our official validation statistics from **Table 13** of the report:

Metric	Value
AUC	0.6775
KS statistic	0.2892
KS optimal score (cutoff)	48.2032
Accuracy Ratio	0.3551

[Point at the ROC curve chart on the left.]

[Trace the teal line with your cursor or laser pointer.]

The ROC curve plots **True Positive Rate** against **False Positive Rate**. The teal line is our scorecard; the dashed diagonal is a random guess. Our curve sits **clearly above the diagonal**, which means the model has **real ranking power** — it is better than flipping a coin.

[Emphasize this result.]

An **AUC of 0.6775** means that if we randomly pick one defaulter and one non-defaulter from the historical data, there is about a **68% chance** the defaulter receives the **lower score**. For a four-predictor model on 500 observations, this is **acceptable academic performance**.

[Point at the Validation Summary panel on the right.]

The **Kolmogorov–Smirnov (KS) statistic** measures the maximum vertical distance between the cumulative distributions of good and bad clients. Our KS is **0.2892**, and the score at which this maximum occurs is **48.2032**.

[Emphasize this result.]

That KS score becomes our **official acceptance threshold**:

Accept if Score ≥ 48.2032 · Reject otherwise

This is a standard industry approach: the cutoff is **data-driven**, not arbitrary.

[Switch to Slide 5 — Portfolio Scoring — Score vs PD (/portfolio)]

[Point at the Score vs PD scatter plot.]

We then apply `score()` and `probdefault()` to the **20 portfolio applicants**. This scatter plot shows each client as a dot: **Score on the x-axis**, **Predicted Probability of Default (PD) on the y-axis**.

[Point at the red vertical dashed line at 48.2032.]

The red vertical line is our **KS cutoff**. Clients to the **right** of the line are accepted; clients to the **left** are rejected.

[Point at green vs red dots.]

Green dots are accepted; red dots are rejected. Notice the relationship is **monotonic**: higher score $\rightarrow$ lower PD. That is exactly what we want from a sound credit scorecard.

[Point at the three risk band cards below.]

We also assign **risk bands** for monitoring:

- **Low Risk**: score $\geq 80 \rightarrow$ **5 accepted clients**
- **Medium Risk**: score 50–80 $\rightarrow$ **9 accepted clients**
- **High Risk**: score $< 50 \rightarrow$ **6 rejected clients**

[Pause for 2 seconds.]

Scores, PDs, and a decision rule are now in place. The following section translates those into **dollar outcomes** — expected loss and minimum interest rates.

Transition $\rightarrow$ Speaker 5

[Brief handoff:]

“Let’s turn to the main numerical results.”

Speaker 5 — Results / Analysis

~4–5 minutes

[Slides 6–8 — Portfolio Results (Tables 15 & 16)]

This section presents the **key numerical results** — who was accepted, total exposure, expected loss, and risk-based minimum interest rates.

[Switch to Slide 6 — Accept vs Reject Decisions (/decisions)]

[Point at the Decision Breakdown bar chart.]

Applying the rule **Score $\geq$ 48.2032** to all 20 applicants:

- **14 clients Accepted**
- **6 clients Rejected**

That is a **70% acceptance rate** — the bank would book 14 loans out of 20 applications.

[Point at the Client Scores vs Threshold bar chart.]

This bar chart shows every client's score sorted from highest to lowest. The red dashed line is the cutoff at **48.20**. You can see clients **4 and 5** at the top with a perfect score of **100**, and client **1** at the bottom with a score of **0**.

[Point at a borderline accepted client — Client 8, score 55.84.]

Client **8** is an interesting borderline case — accepted with a score of **55.84**, just above the threshold, but classified as **Medium Risk** because the score is below 80.

[Scroll to the portfolio results table.]

[Point at rows for clients 4, 5, and 1.]

The full table matches **Table 15** in our report. For example:

- Client **4**: Score **100**, PD **12.79%**, Accepted, Expected Loss **\$5,114**
- Client **1**: Score **0**, PD **70.59%**, Rejected, Expected Loss **\$28,234** (*not booked*)

[Switch to Slide 7 — Expected Loss Dashboard (/expected-loss)]

[Point at the three summary metric cards.]

For the **14 accepted clients only**, using **EL = PD $\times$ LGD $\times$ EAD**:

Metric	Value
Total Expected Loss	\$129,329.63
Average PD (accepted)	23.09%
Accepted Exposure	\$1,400,000

[Point at the Expected Loss bar chart.]

Each bar is one accepted client's expected loss in dollars. Taller bars mean more capital at risk. Client **8** has the highest EL among accepted clients — about **\$13,402** — because they have the highest PD among those still above the cutoff.

[Emphasize this result.]

The **\$129,329.63** total is the dollar amount the bank should **provision or cover through pricing** on the accepted book, assuming a **60% recovery rate**.

[Switch to Slide 8 — Risk-Based Interest Rates (/pricing)]

[Point at the minimum interest rate bar chart.]

For pricing, we use the assignment formula:

$$r_{\min} = PD \times LGD \text{ (no separate base rate)}$$

[Point at the gold dashed average line.]

The **average minimum break-even rate** across the 14 accepted clients is **9.24%**. The safest accepted clients (IDs **4 and 5**) need only **5.11%** to break even on expected loss. The highest accepted rate is **13.40%** for client **8**.

[Point at the two summary boxes below the chart.]

The **lowest accepted rate is 5.11%**; the **highest rejected rate is 28.23%** (client **1**). Rejected clients would require prohibitively high rates — another reason rejection makes business sense.

[Optional: Slide 12 — MATLAB & Excel Source Charts (/outputs/matlab) — expected_loss_per_client.png and min_rate_per_client.png]

These MATLAB-generated figures in our appendix match what you see in the dashboard.

[Pause for 2 seconds.]

The quantitative story is complete. The final section discusses **what these results mean in practice**, acknowledges **limitations**, and closes with a summary.

Transition → Speaker 6

[Brief handoff:]

“To close: insights, limitations, and conclusion.”

Speaker 6 — Discussion / Insights + Conclusion

~4–5 minutes

[Slides 9–11 — Discussion & Conclusion]

To close, this section covers **business insights**, **limitations**, and the **final portfolio recommendation**.

[Switch to Slide 9 — Model Performance Summary (/performance)]

[Point at the Validation Radar chart.]

Looking back at model performance holistically: our scorecard achieves **AUC 0.6775**, **KS 0.2892**, and an **Accuracy Ratio of 0.3551**. These metrics tell us the model **ranks clients meaningfully**, even though it is not perfect — no real-world scorecard with only four variables ever is.

[Point at the Project Outcomes list on the right.]

Discussion — What does this mean for the bank?

1. **Profitable selection:** By rejecting the 6 highest-risk applicants, the bank avoids clients with PDs above **44%** and expected losses up to **\$28,234** per \$100,000 loan.
2. **Risk-based pricing:** Instead of charging everyone the same rate, the bank can price from **5.11%** for the safest accepted clients to **13.40%** for borderline cases — aligning price with risk.
3. **Capital planning:** The **\$129,329.63** expected loss figure gives management a concrete number for **provisions and stress testing** on the \$1.4 million book.
4. **Transparency:** Every decision traces back to observable characteristics and a documented cutoff — important for regulators and internal audit.

[Switch to Slide 10 — Final Portfolio Decision (/final)]

[Point at the headline banner — 14 clients, \$1.4M exposure.]

Our **final recommended portfolio** is **14 clients**, **\$1.4 million** in exposure, **\$129,329.63** in expected loss, with an average minimum rate of **9.24%**.

[Point at the accepted vs rejected client lists.]

Five clients are **Low Risk** (score above 80). Nine are **Medium Risk** (between 50 and 80). All six rejected clients fall in the **High Risk** band (score below 50).

[Switch to Slide 11 — Live Client Test (/test) — optional 30-second demo]

[Select Client 4 from the dropdown. Click “Run credit test.”]

As a quick demo: if we test **Client 4** live in RiskLens, the dashboard returns the exact report values — Score **100**, PD **12.79%**, Accepted, EL **\$5,114**, minimum rate **5.11%**. This confirms our dashboard and MATLAB pipeline are aligned.

[Pause for 2 seconds.]

Limitations — intellectual honesty matters in FinTech:

- We use only **four predictors**. Real banks typically use dozens — credit bureau scores, debt ratios, industry codes, and more.
- **AUC of 0.6775** is reasonable for a four-variable model but would need improvement before production deployment.
- We assume a **flat \$100,000 loan** and a **fixed 60% recovery rate** for all clients — real portfolios are more granular.
- The model is trained on **historical data**; economic shocks can shift default patterns over time — **monitoring and recalibration** would be required.

[Emphasize this result.]

Despite these limits, the project demonstrates the **full end-to-end credit scorecard lifecycle**: from raw Excel data through WOE binning, logistic regression, validation, portfolio decision, expected loss, and risk-based pricing.

[Return to title slide, or remain on Slide 10 — Final Portfolio Decision (/final)]

Conclusion

To summarize in three sentences:

1. We built a **MATLAB credit scorecard** on **500 historical clients** with **autobinning**, achieving **AUC 0.6775**.
2. Applied to **20 applicants**, we **accepted 14** at a KS cutoff of **48.2032**, with **\$129,329.63** total expected loss on **\$1.4 million** exposure.
3. Minimum interest rates range from **5.11%** to **13.40%** on the accepted book, averaging **9.24%**.

[Pause for 2 seconds. Speak slowly and clearly.]

Thank you for your attention — we're happy to take your **questions**.

[Pause. Remain at the front for Q&A.]

Q&A preparation (all members)

Quick reference for likely questions:

Question	Short answer
Why no train/test split?	Assignment slide: train on all 500; validate on HistoricalData.
Why KS cutoff and not 50% PD?	KS maximizes separation between good/bad distributions on the score.
Why is Income the strongest predictor?	IV = 0.2355; lower income → higher historical default rate.
What is WOE?	Log ratio measuring bin's good/bad separation; used in logistic regression.
Why is total EL \$129,329.63 but some table ELs are rounded?	Table 15 shows rounded dollars; Table 16 total is MATLAB's precise sum.
What is LGD?	40% — the loss per dollar lent if default occurs (1 – 60% recovery).
Can we see the MATLAB code?	Yes — <code>FinTech_Scorecard_Project.m</code> in the repo and <code>Final/</code> submission.

Rehearsal checklist

- ☐ Dashboard running locally or deployed URL tested
- ☐ Each speaker rehearsed with a timer (target: 4:00–5:00 per person)
- ☐ Transitions practiced aloud
- ☐ Backup: PDF report open on second device
- ☐ Everyone can answer at least one Q&A item
- ☐ Screen font size readable from the back of the room

Script aligned with `Final/fintech (1).pdf`, `RiskLens` navbar steps 1–12, and `fintech-dashboard/src/config/presentationNav.ts`. Last updated: May 2026.